Factor Validation of Activities

We examine the empirical correlates of everyday cultural practice by analyzing past-year participation frequencies across 20 distinct activities spanning institutional highbrow arts, commercial entertainment, domestic tasks, and solitary leisure routines. Because these responses are recorded on an ordinal seven-point frequency scale, we estimate principal components on the polychoric correlation matrix using varimax rotation to establish a preliminary exploratory assignment of items to factors. The three-component solution accounts for 63.2 percent of total variance, supported by a Kaiser-Meyer-Olkin sampling adequacy metric of .955 (Table 1).[1]

Substantively, the orthogonal solution uncovers a structural division between public, institutionalized cultural practices and private, domestic routines. Fourteen items coalesce into an expansive public arts factor, uniting traditional fine arts (symphonies, art galleries, theatrical performances) with broadly attended public events (festivals, movies, sporting events). Three items—taking walks, regular exercise or yoga, and literary reading—cluster into an independent solitary leisure dimension. Three remaining practices (gardening, home or automobile repair, and dining at fast-food restaurants) define a residual practical-domestic dimension.[2]

We evaluate the target structure against more parsimonious alternatives using confirmatory factor analysis with diagonally weighted least squares (WLSMV estimation, N = 1,243; Table 2), which relaxes the varimax orthogonality assumption by allowing the factors to be correlated. The three-factor model outperforms more parsimonious alternatives. Forcing residual practical items onto the arts factor produces a significant decrement in fit (scaled $\Delta\chi^2(2) = 121.6$, $p < .001$), as does forcing them onto solitary leisure ($\Delta\chi^2(2) = 51.0$, $p < .001$). A single-factor model specifying a generalized omnivorousness dimension performs worst of all ($\Delta\chi^2(3) = 146.4$, $p < .001$). Although the three-factor confirmatory model fits better than any rival (CFI = .956, RMSEA = .095, SRMR = .061), the elevated RMSEA reflects the restrictive requirement that each practice load onto only a single factor. Everyday cultural activities naturally cross domain boundaries in ways that strict zero-loading constraints penalize.

Because everyday cultural practices rarely divide into mutually exclusive silos, we estimate an exploratory structural equation model (ESEM) using target rotation. By anchoring primary loadings to the varimax configuration while permitting items to exhibit small, substantively

[1] Complete responses on all 20 items are available for 1,243 respondents, who form the analytic sample for all measurement models.

[2] We also estimated an exploratory factor analysis with direct oblimin rotation on the same polychoric matrix (Table A1). The oblique model fits the 20 items adequately ($\chi^2(133) = 290.8$, $p < .001$, TLI = .911, RMSEA = .080 [.076, .084]). Under direct oblimin, the public arts factor remains intact, but the secondary dimensions lose coherence. Assigning items based on highest loadings shifts literary reading, gardening, and home repair onto the arts factor, isolating fast food as a single-item residual. Forcing that assignment into confirmatory models yields inadmissible solutions characterized by negative residual variances and latent correlations exceeding unity. Because the oblique factor correlations in the exploratory solution are small (.19 between arts and leisure, .15 between arts and residual, and .01 between leisure and residual), the orthogonal and oblique orientations differ minimally in practice, and nothing in the data privileges one rotation over the other.

plausible cross-loadings, the ESEM achieves a good fit to the observed polychoric covariances (CFI = .979, RMSEA = .074 [.070, .079], SRMR = .035), improving over the confirmatory baseline (scaled $\Delta\chi^2(34) = 746.7$, $p < .001$; Table 2).[3] Latent factor correlations confirm that while these three domains are distinct, they are modestly interrelated; public arts correlates positively with solitary leisure ($r = .36$) and residual domestic practices ($r = .21$), while leisure and residual practices correlate only weakly ($r = .11$; Table 3). As Table 3 shows, all fourteen public arts items maintain primary loadings between .47 and .97. Solitary leisure is anchored firmly by walking (.78) and physical exercise (.42), though exercise also displays a secondary affinity with public cultural engagement (.32). Literary reading functions as the peripheral anchor of this leisure factor across all specifications (.44 in Table 1, .21 in Table A1, and .19 in Table 3), a structural attenuation that proves central to how solitary leisure behaves across alternative operationalizations. Domestic tasks (gardening at .45, home or auto repair at .32) and fast food avoidance (−.38) define the residual factor, which also absorbs modest secondary loadings from institutional excursions (museums at .24, historic sites at .23) and outdoor recreation (hiking at .21).[4]

[3]Geomin rotation yields identical omnibus fit because rotation does not alter the implied covariance matrix, but it redistributes factor content in an uninterpretable fashion. Under geomin, commercial entertainment (movie theaters, sporting events, dining out) combines with fast food to form the second factor, displacing walking to the third dimension. We retain the varimax targets because they align with the theoretical design of the battery and preserve substantive interpretability.

[4]Two measurement caveats qualify this three-factor solution. First, the residual practical-domestic factor exhibits modest internal coherence (McDonald's $\omega = .55$, ordinal $\alpha = .31$), contrasting with the high reliability of public arts ($\alpha = .96$) and solitary leisure ($\alpha = .49$). Second, several highbrow practices have sparse counts in the highest frequency tier (two to nine respondents for symphonies, dance performances, and carnivals); sensitivity analyses collapsing the top two categories across these items leave factor assignments and fit indices unchanged.

Table 1. Varimax Loadings for the Activity Battery

Item	Assigned target	Arts	Leisure	Residual
Museum	Arts	0.80	0.21	0.21
Art gallery	Arts	0.83	0.19	0.20
Symphony, orchestra, or opera	Arts	0.89	0.06	0.23
Carnival, fair, or amusement park	Arts	0.83	0.13	0.01
Music concert or festival	Arts	0.84	0.17	0.02
Play or musical	Arts	0.84	0.09	0.13
Library	Arts	0.49	0.39	0.15
Fancy restaurant	Arts	0.72	0.26	−0.15
Dance performance	Arts	0.89	0.00	0.15
Movie theater	Arts	0.74	0.23	−0.24
Attend sports	Arts	0.79	0.21	−0.07
Hiking, camping, or boating	Arts	0.52	0.37	0.32
Historic site	Arts	0.70	0.23	0.26
Go to festival	Arts	0.78	0.27	0.15
Go for a walk	Leisure	−0.14	0.87	0.07
Exercise or yoga	Leisure	0.35	0.60	−0.19
Read a novel, poem, or play	Leisure	0.27	0.44	0.18
Gardening	Residual	0.27	0.31	0.62
Fast food	Residual	0.21	0.22	−0.66
Home auto repair	Residual	0.38	0.24	0.49

Note. Varimax-rotated component loadings from a principal components analysis (N = 1,243). The assigned target column reports the factor on which each item is assigned in the ESEM targets. The three components together explain 63.2 percent of the variance.

Table 2. Confirmatory and Exploratory Structural Models of the Activity Battery

Model	df	Scaled χ^2	CFI	RMSEA	SRMR	Scaled $\Delta\chi^2$ (df) vs. three-factor CFA
One-factor CFA	170	2225.6	.952	.099	.064	146.4 (3), $p < .001$
Two-factor CFA, residual merged into arts	169	2199.3	.952	.098	.063	121.6 (2), $p < .001$
Two-factor CFA, residual merged into leisure	169	2120.8	.954	.096	.062	51.0 (2), $p < .001$
Three-factor CFA	167	2051.2	.956	.095	.061	—
ESEM, target rotation	133	1045.1	.979	.074	.035	746.7 (34), $p < .001$

Note. N = 1,243. WLSMV estimation. The three-factor CFA is nested within the ESEM's implied covariance class as the case with zero cross-loadings, so the ESEM row reports a scaled difference test against it. The ESEM degrees of freedom reflect its rotation constraints. Latent factor correlations in the three-factor CFA are .76 (arts and leisure), .68 (arts and residual), and .59 (leisure and residual). RMSEA 90% confidence intervals are [.095, .102] (one-factor), [.095, .102] (residual into arts), [.093, .100] (residual into leisure), [.092, .099] (three-factor), and [.070, .079] (ESEM). Scaled difference tests use the Satorra-Bentler scaled statistic with per-model scaling factors.

Table 3. ESEM Loadings with Target Rotation

Item	Target factor	Arts	Leisure	Residual
Museum	Arts	0.78	0.03	0.24
Art gallery	Arts	0.83	−0.02	0.20
Symphony, orchestra, or opera	Arts	0.95	−0.22	0.11
Carnival, fair, or amusement park	Arts	0.82	0.04	−0.11
Music concert or festival	Arts	0.86	0.02	−0.11
Play or musical	Arts	0.90	−0.16	0.02
Library	Arts	0.47	0.16	0.14
Fancy restaurant	Arts	0.71	0.15	−0.20
Dance performance	Arts	0.97	−0.25	0.01
Movie theater	Arts	0.75	0.11	−0.30
Attend sports	Arts	0.80	0.08	−0.21
Hiking, camping, or boating	Arts	0.48	0.23	0.21
Historic site	Arts	0.65	0.11	0.23
Go to festival	Arts	0.78	0.10	0.04
Go for a walk	Leisure	−0.22	0.78	0.12
Exercise or yoga	Leisure	0.32	0.42	−0.17
Read a novel, poem, or play	Leisure	0.27	0.19	0.16
Gardening	Residual	0.23	0.17	0.45
Fast food	Residual	0.20	0.14	−0.38
Home auto repair	Residual	0.34	0.14	0.32

Note. Standardized loadings from an ESEM with target rotation (WLSMV, N = 1,243). Factor correlations: arts and leisure .36, arts and residual .21, leisure and residual .11.

Factor Validation of Network Composition

The position generator batteries assess personal networks across sixteen social categories, measured separately for casual acquaintances (weak ties) and close confidants (strong ties).[5] Rather than collapsing social ties into an omnibus index of network diversity, the measurement model uncovers three distinct relational structures across both relational spheres: a generalized breadth of ties spanning ethnic, geographic, and socioeconomic boundaries (Network Variety), and two distinct channels of partisan homophily (Liberal Composition and Conservative Composition). This separation confirms that demographic bridging and ideo-

[5]Gender items (knowing men, knowing women) are excluded because they display marked structural ambiguity across both relational domains. In exploratory analyses, knowing women splits nearly equally between the liberal and conservative factors in weak networks (.59 and .60), while knowing men loads heavily on both factors in weak ties (.68 and .57) and on conservative composition in strong ties (.62, alongside a .58 loading on liberal composition). In confirmatory models, retaining them lowers fit to CFI = .896 and RMSEA = .116 for weak ties and CFI = .918 and RMSEA = .105 for strong ties. Re-estimating regression models without the four gender items leaves every substantive finding and significance verdict intact, with no network coefficient shifting by more than .049.

logical sorting operate as distinct relational dynamics; maintaining connections across varied social strata does not imply alignment with a particular political camp.

In everyday American life, however, social diversity and political sorting rarely occupy hermetically sealed environments. Conventional confirmatory factor models, which impose strict zero-loading restrictions by forcing each social position onto a single factor, fit the data poorly and mischaracterize everyday network topology (Table A2).[6] By penalizing the natural secondary affinities that exist among demographic and political identities, strict confirmatory models artificially force shared social spaces into the latent factor correlations, inflating the overlap between network variety and liberal composition to over .80 and inducing severe collinearity. Estimating exploratory structural equation models (ESEM) with target rotation resolves this distortion (Table 4). By permitting positions to display modest, substantively meaningful secondary affinities, the ESEM achieves an excellent fit (CFI $\geq$ .975, RMSEA $\leq$ .069, SRMR $\leq$ .033), showing that while demographic diversity and progressive sorting intersect in everyday networks, they remain distinct, moderately related dimensions ($r \approx .50$).

The pattern of secondary loadings in Table 4 reveals a meaningful shift in how racial diversity and political sorting operate across levels of intimacy. In casual acquaintance networks, minoritized racial and ethnic ties (Black and Hispanic or Latine acquaintances) straddle both network variety and progressive political composition. This pattern reflects the institutional organization of everyday social life; for many Americans, casual contact across racial lines occurs within progressive urban, educational, or professional environments. In close personal networks, by contrast, these same ties anchor structural variety almost exclusively. When Americans form close, confiding bonds across racial lines, those ties represent genuine cross-boundary social bridging rather than incidental proximity to progressive politics. Conservative composition, meanwhile, exhibits a starkly consolidated relational boundary across both weak and strong spheres; it is anchored exclusively by church attendance, political conservatism, rural residence, and White ties. White ties also display secondary affinities with liberal networks, capturing the partisan divide within the majority population, whereas conservative homophily remains structurally insular and segregated from ethnic diversity.

All three network dimensions achieve high internal reliability across both tie types, with model-implied omegas ranging from .88 to .94 (Table 4). In the structural equation models that follow, we use the simple-structure confirmatory factors as our primary operationalization to maintain clean, non-overlapping composites (Table A2), while using the ESEM specifications as a robustness benchmark to verify that our substantive findings do not depend on zero-loading restrictions.[7]

[6] In simple-structure confirmatory models, freeing residual covariances between polar opposites (very liberal with very conservative, little church with lots of church, and city with rural) modestly improves fit (Table A2). Under the target-rotated ESEM, however, secondary cross-loadings absorb this bipolarity, rendering those residual covariances redundant and producing numerical instability in strong-tie estimation. We therefore retain the target ESEM without residual covariances.

[7] We do not estimate a joint model that links the weak and strong factors. Such a model requires cross-tie factor correlations near one for the variety factors, and it was not positive definite under either the confirmatory or the target-rotated specification. The separate models are therefore the measurement models for both tie types.

Table 4. Standardized Loadings from the Target-Rotated ESEM of the Weak-Tie and Strong-Tie Batteries

	Weak ties			Strong ties		
Item	Variety	Liberal	Conservative	Variety	Liberal	Conservative
Asian people	**0.61**	0.33	−0.03	**0.73**	0.15	−0.06
Second home	**0.50**	−0.03	0.34	**0.56**	−0.09	0.31
Born elsewhere	**0.48**	0.38	0.02	**0.70**	0.19	−0.04
Middle Eastern or North African people	**0.77**	0.15	0.00	**0.88**	0.02	−0.02
Native Hawaiian or Pacific Islander people	**0.93**	−0.04	−0.06	**0.93**	−0.08	−0.05
American Indian people	**0.84**	−0.12	0.06	**0.85**	−0.14	0.09
LGBTQ people	0.03	**0.80**	−0.07	0.24	**0.65**	−0.10
Little church	−0.11	**0.58**	0.32	−0.07	**0.54**	0.36
Very liberal	−0.02	**0.78**	0.05	0.08	**0.81**	−0.01
Hispanic or Latine people	0.32	**0.54**	0.01	**0.59**	0.27	−0.04
Black people	0.36	**0.36**	0.16	**0.55**	0.18	0.15
City people	0.30	**0.44**	0.07	0.35	**0.36**	0.12
Lots of church	0.22	−0.02	**0.67**	0.29	−0.09	**0.63**
Very conservative	0.11	−0.20	**0.90**	0.14	−0.20	**0.81**
White people	−0.19	0.41	**0.68**	−0.15	0.36	**0.72**
Rural people	−0.07	0.11	**0.61**	−0.05	0.16	**0.60**

Note. Standardized loadings from exploratory structural equation models with target rotation (WLSMV, polychoric; N = 1,258 per battery). Bold entries are the primary loadings from the target, and all other loadings are estimated rather than fixed at zero. Gender items are excluded. Factor correlations are .52 (variety and liberal), .46 (variety and conservative), and .47 (liberal and conservative) for weak ties, and .48, .54, and .34 in the same order for strong ties. Item labels are abbreviated from the variable names.

Structural Models of Activity Participation

To evaluate how network variety and ideological composition relate to cultural participation, we estimate structural equation models separately across weak and strong ties on the confirmatory network and activity factors defined above.[8] Each system incorporates the three validated network dimensions (variety, liberal composition, and conservative composition), the three activity domains (public arts, solitary leisure, and practical-domestic residual), and

[8]Several methodological caveats qualify these structural specifications. First, because these survey data are cross-sectional, the models identify patterns of relational association rather than causal trajectories, illuminating how social ties channel demographic advantages without assuming networks are exogenous to cultural dispositions. Second, estimating separate systems for weak and strong ties preserves empirical identification at the expense of modeling cross-tie interdependencies simultaneously. Third, the practical-domestic residual factor exhibits modest reliability ($\omega = .55$), so its structural paths should be interpreted with caution.

sociodemographic predictors. Gender, race and ethnicity, and childhood arts exposure enter as specified in our regression models; age enters linearly in years. Education is specified as a single ordinal scale ranging from high school or less (1) to graduate credentials (4), capturing monotonic human-capital accumulation across adjacent educational tiers. Household income and political conservatism (ranging from 1 for very liberal to 7 for very conservative) enter as linear predictors. Estimated on complete cases (N = 1,227) using weighted least squares with mean and variance adjustment (WLSMV), both models converge cleanly, yield positive definite covariance matrices, and fit the data well (CFI = .906, RMSEA = .055 [.053, .057], SRMR = .061 for weak ties; CFI = .903, RMSEA = .056 [.054, .058], SRMR = .059 for strong ties).

The structural estimates reveal that network variety functions as the central relational channel for public arts attendance across both weak ($\beta = .85$, $p < .001$) and strong ($\beta = .68$, $p < .001$) ties (Table 5). Once network variety is held constant, liberal network composition exhibits a significant negative net association with public arts participation ($-.58$, $p < .001$ for weak ties; $-.27$, $p < .001$ for strong ties). Conservative composition, by contrast, shows no discernible relationship with public arts engagement in either relational sphere. Wald tests of joint significance confirm that network factors exert a collective influence on arts participation ($\chi^2(3) = 137.1$ for weak ties; $\chi^2(3) = 207.3$ for strong ties, both $p < .001$), with a parallel configuration emerging for practical domestic activities.[9]

Solitary leisure follows a different pattern, which marks the limits of network influence. Unlike public arts attendance, which requires social coordination and carries public signaling value, solitary routines such as walking, reading, and personal fitness unfold in private settings. Consequently, network variety plays no role in driving solitary leisure ($\beta = .06$ for weak ties; $\beta = -.01$ for strong ties). Only liberal composition displays a modest positive association with solitary routines ($\beta = .26$, $p = .10$ for weak ties; $\beta = .26$, $p = .02$ for strong ties), and block tests indicate joint network associations despite the weakness of the individual paths.

The substantial overlap between weak-tie variety and liberal composition warrants careful substantive interpretation. After conditioning on covariates, these two dimensions correlate at .83, yielding variance inflation factors of 3.4 for variety and 5.5 for liberal composition. In bivariate specifications that omit variety, the association between liberal composition and arts attendance is positive; it turns negative only when estimated jointly with network breadth. The resulting suppression dynamic clarifies a central theoretical puzzle; in contemporary American society, demographic diversity and progressive political environments frequently intersect. Bivariate models misattribute the cultural benefits of demographic bridging to progressive homophily. When structural variety is held constant, however, an insular concentration of progressive ties does not expand cultural practice—it constrains it. Broad network bridges across diverse social positions, rather than ideological sorting into progressive enclaves, provide the relational capital that sustains public cultural engagement.

[9]Sensitivity checks confirm the stability of these structural estimates. Omitting conservative composition from the arts equation alters the remaining coefficients only modestly, and freeing the three antonym residual covariances in the confirmatory network models (Table A2) shifts standardized network paths by at most .03, leaving every substantive pattern and significance threshold unchanged.

Conversely, holding liberal composition constant reveals the full power of structural variety, elevating its coefficient from .36 in isolation to 1.08 in joint estimation.

Sociodemographic patterns reinforce these relational dynamics across both models (Tables 6–9). Educational attainment predicts public arts attendance, solitary leisure, and practical domestic tasks, while expanding network variety across weak ($\beta = .29$, $p < .001$) and strong ($\beta = .27$, $p < .001$) ties. Asian American and Pacific Islander respondents report lower public arts participation relative to White respondents in both models ($\beta = -.12$ weak, $-.10$ strong, both $p < .001$). Minoritized racial and ethnic groups (Black, Hispanic or Latine, AAPI, and multiracial respondents) consistently maintain fewer conservative ties than White respondents. Political conservatism predicts conservative network sorting and depresses liberal ties across both relational domains.

We also re-estimated the network-to-activity paths with the target-rotated ESEM blocks in place of the confirmatory network factors, keeping the activity factors and covariates unchanged.[10] The weak-tie variety and liberal paths keep their direction and significance, at .55 and $-.36$ in standardized form, against .85 and $-.58$ under the confirmatory factors, and the weak-tie conservative path stays null. The weak-tie joint test for solitary leisure remains significant ($\chi^2(3) = 15.9$, p = .001). The strong-tie variety path remains significant at .58 (p = .005). The strong-tie liberal path on arts shrinks to $-.24$ and is not significant (p = .65), and the strong-tie joint test for solitary leisure falls to $\chi^2(3) = 6.4$, p = .10. The weak-tie composition effects are therefore more modest than the confirmatory estimates suggest, and the strong-tie liberal coefficient should not be read as a firm finding. The residual factor paths are null under both measurements.

Table 5. Standardized Network-to-Activity Paths from the Separate Weak-Tie and Strong-Tie Structural Models

Network factor	Arts	Solitary leisure	Residual
Weak variety	0.85***	0.06	0.86***
Weak liberal	−0.58***	0.26	−0.89***
Weak conservative	0.09	−0.07	0.47***
Strong variety	0.68***	−0.01	0.47***
Strong liberal	−0.27***	0.26*	−0.33**
Strong conservative	−0.02	0.01	0.23*

Note. Standardized coefficients from separate WLSMV structural models for weak ties and strong ties (N = 1,227 each). * p < .05, ** p < .01, *** p < .001. Each model includes gender, education (single ordinal variable), childhood arts, income, age in years, race and ethnicity, and political ideology as predictors of all network and activity factors. The weak-tie liberal and variety coefficients on arts are estimated jointly and should be read as conditional on

[10]We do not report covariate paths onto the ESEM blocks. In lavaan 0.7-3, regressions of covariates onto target-rotated factors produced coefficients whose signs contradicted the item-level associations, and changing the exogeneity specification did not resolve the problem. The network-to-activity estimates reported here come from models that include the same covariates.

each other.

Table 6. Sociodemographic Predictors of Weak-Tie Network Composition Factors

Predictor	Variety	Liberal	Conservative
Age (years)	−0.11***	−0.04	0.00
Education (ordinal, 1 to 4)	0.29***	0.13***	0.14***
Childhood arts exposure	0.26***	0.17***	0.10**
Household income	0.09**	0.07*	0.10**
Political ideology (1 to 7)	0.03	−0.27***	0.26***
Man (ref. woman)	0.00	−0.07*	−0.09**
Non-binary/trans (ref. woman)	−0.04	−0.05	−0.08**
Black (ref. White)	0.05	0.03	−0.15***
Hispanic or Latine (ref. White)	0.05	0.06*	−0.14***
AAPI (ref. White)	0.07**	−0.02	−0.16***
Mixed/other (ref. White)	0.06*	0.04	−0.11***

Note. Standardized coefficients from the weak-tie structural model (N = 1,227). * $p < .05$, ** $p < .01$, *** $p < .001$. Reference categories are shown in parentheses. Education is a single ordinal variable. Political ideology ranges from 1 (very liberal) to 7 (very conservative).

Table 7. Sociodemographic Predictors of Strong-Tie Network Composition Factors

Predictor	Variety	Liberal	Conservative
Age (years)	−0.11***	−0.03	−0.01
Education (ordinal, 1 to 4)	0.27***	0.18***	0.17***
Childhood arts exposure	0.29***	0.22***	0.13***
Household income	0.01	0.09**	0.03
Political ideology (1 to 7)	0.07*	−0.32***	0.35***
Man (ref. woman)	−0.01	−0.08**	−0.09**
Non-binary/trans (ref. woman)	−0.03	−0.02	−0.05
Black (ref. White)	0.12***	−0.02	−0.14***
Hispanic or Latine (ref. White)	0.08**	0.00	−0.11***
AAPI (ref. White)	0.02	−0.08**	−0.21***
Mixed/other (ref. White)	0.04	−0.04	−0.16***

Note. Standardized coefficients from the strong-tie structural model (N = 1,227). * $p < .05$, ** $p < .01$, *** $p < .001$. Reference categories are shown in parentheses. Education is a single ordinal variable. Political ideology ranges from 1 (very liberal) to 7 (very conservative).

Table 8. Sociodemographic Predictors of Weak-Tie Activity Factors

Predictor	Arts	Solitary leisure	Residual
Age (years)	−0.08**	−0.15***	0.14**
Education (ordinal, 1 to 4)	0.20***	0.32***	0.06
Childhood arts exposure	0.16***	0.16***	0.07
Household income	0.04	0.11**	−0.02
Political ideology (1 to 7)	−0.12*	−0.01	−0.27**
Man (ref. woman)	−0.04	−0.10*	−0.01
Non-binary/trans (ref. woman)	0.02	−0.02	0.03
Black (ref. White)	0.04	−0.03	−0.01
Hispanic or Latine (ref. White)	0.03	−0.04	0.05
AAPI (ref. White)	−0.12***	−0.07	−0.02
Mixed/other (ref. White)	−0.05	0.00	0.07

Note. Standardized coefficients from the weak-tie structural model in Table 6 (N = 1,227). * p < .05, ** p < .01, *** p < .001. The activity equations also include the three weak-tie network factors, whose coefficients are reported in Table 5.

Table 9. Sociodemographic Predictors of Strong-Tie Activity Factors

Predictor	Arts	Solitary leisure	Residual
Age (years)	−0.09***	−0.17***	0.12**
Education (ordinal, 1 to 4)	0.25***	0.32***	0.16***
Childhood arts exposure	0.16***	0.16***	0.10*
Household income	0.10***	0.11*	0.07
Political ideology (1 to 7)	−0.04	−0.01	−0.10
Man (ref. woman)	−0.02	−0.09*	0.01
Non-binary/trans (ref. woman)	0.02	−0.03	0.02
Black (ref. White)	−0.04	0.00	−0.09*
Hispanic or Latine (ref. White)	−0.03	−0.01	−0.04
AAPI (ref. White)	−0.10***	−0.04	0.00
Mixed/other (ref. White)	−0.07**	0.03	0.04

Note. Standardized coefficients from the strong-tie structural model in Table 7 (N = 1,227). * p < .05, ** p < .01, *** p < .001. The activity equations also include the three strong-tie network factors, whose coefficients are reported in Table 5.

Decomposing total associations into direct and network-mediated pathways reveals how stratification advantages and political dispositions are translated into cultural practice (Tables 11 and 12; Figures 5–10). Educational attainment exhibits the strongest total association with public arts participation across both relational models ($\beta = .38$, $p < .001$). In weak networks, nearly half of this educational advantage (network-mediated effect = .18, $p < .001$) is channeled through social networks, driven primarily by network variety (.25) and partially counterbalanced by liberal composition (−.07). In strong networks, the network-mediated

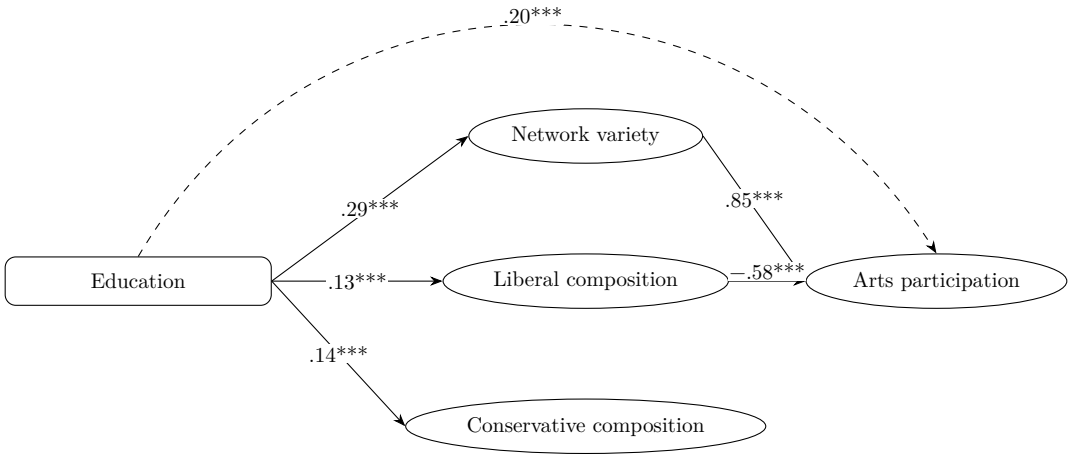
share accounts for roughly one-third of the total association (.13, $p < .001$), operating through network variety (.18) and liberal composition (−.05). By contrast, the educational gradient for solitary leisure operates almost entirely as a direct effect ($\beta = .32$, $p < .001$ in weak ties), with network mediation playing virtually no role (.04, not significant). In strong networks, education exerts a modest indirect path on leisure through liberal composition (.05, $p = .04$). For practical domestic activities, education operates predominantly through network pathways (.20 in weak ties; .11 in strong ties), fueled by network variety (.25 and .13) and tempered by liberal composition (−.11 and −.06), with a modest positive contribution from conservative ties in weak networks (.07).

Childhood arts socialization displays a parallel dual pathway structure. Early exposure to the arts yields a total association of .29 ($p < .001$) with adult arts attendance across both tie types. Approximately 45 percent of this total effect (network-mediated effect = .13, $p < .001$) operates indirectly through adult network formation, almost exclusively through the construction of diverse network ties (.22 for weak ties; .19 for strong ties), offset slightly by liberal composition (−.10 and −.06). The direct socialization effect remains substantial ($\beta = .16$, $p < .001$). Childhood socialization also exerts significant network-mediated effects on solitary leisure (.06 across both tie types) and on practical domestic tasks (.12 in weak ties; .09 in strong ties). In weak networks, the indirect leisure effect reflects the joint precision of negatively covarying path estimates rather than a single dominant channel; in strong networks, it is mediated specifically by liberal composition (.06, $p = .03$).

Political conservatism exhibits a suppression dynamic. The total association between conservatism and public arts attendance is modest and positive (.09, $p < .001$ across both tie types), but this net estimate conceals opposing direct and indirect forces. In weak ties, conservative respondents exhibit a significant negative direct orientation toward public arts participation (−.12, $p < .05$). That negative direct inclination is more than offset by a substantial positive network-mediated effect (.20, $p < .001$), flowing primarily through the liberal composition channel (.16), with a small contribution from network variety (.02). Because conservative individuals maintain fewer liberal acquaintances, they bypass the negative net constraint that progressive network insularity imposes on arts participation. In strong ties, the direct association is negligible (−.04, not significant), while the network-mediated path remains positive (.13, $p < .001$). A similar dynamic governs practical domestic activities in weak networks, where a negative direct association (−.27, $p < .01$) is overridden by positive network mediation (.39, $p < .001$) running through both liberal (.24) and conservative (.12) composition channels. For solitary leisure, political conservatism displays a negative total association (−.10 in weak ties; −.09 in strong ties), but aggregate network-mediated effects do not reach significance in either domain, despite a modest negative indirect path through strong-tie liberal composition (−.08, $p = .03$).

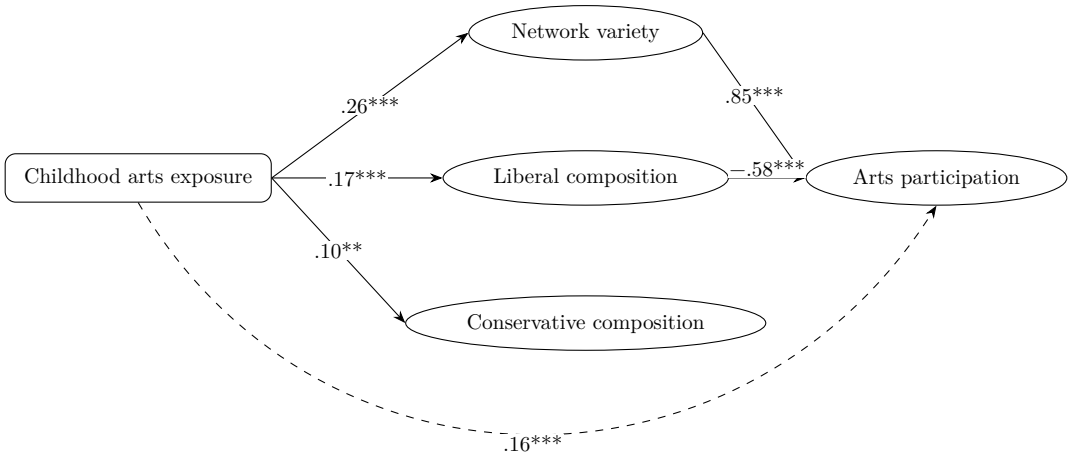
The path decompositions illuminate the relational pathways through which stratification shapes cultural practice, under standard cross-sectional mediation assumptions that condition on our observed demographic covariates. The network-mediated effects quantify the proportion of total demographic associations channeled through social network configurations, clarifying the relational mechanisms that link social origins to everyday cultural lifestyles.

Figure 5. Paths from Education to Arts Participation, Weak Ties



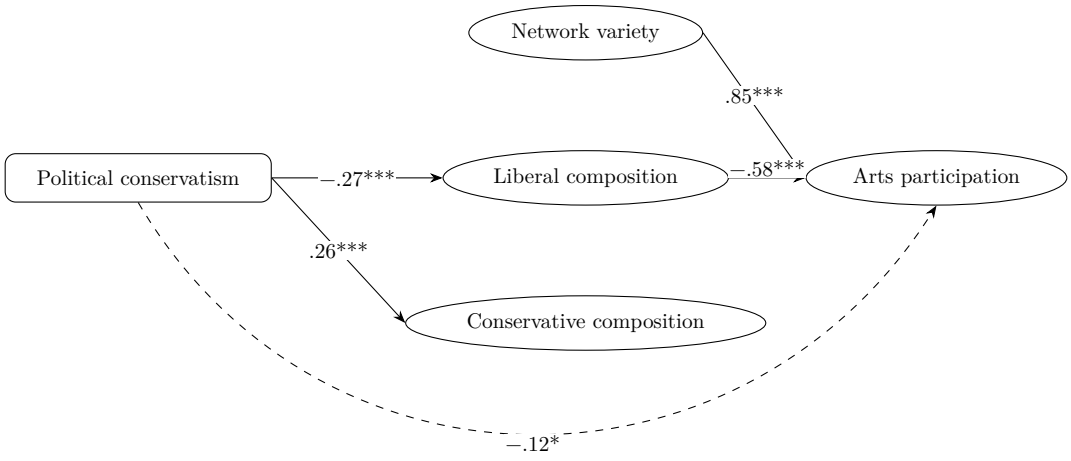
Note. Standardized coefficients from the weak-tie structural model (N = 1,227). Solid arrows are paths through the network factors, dashed arrows are direct paths, and only paths significant at $p < .05$ are drawn. Asterisks follow Table 5.

Figure 6. Paths from Childhood Arts Exposure to Arts Participation, Weak Ties



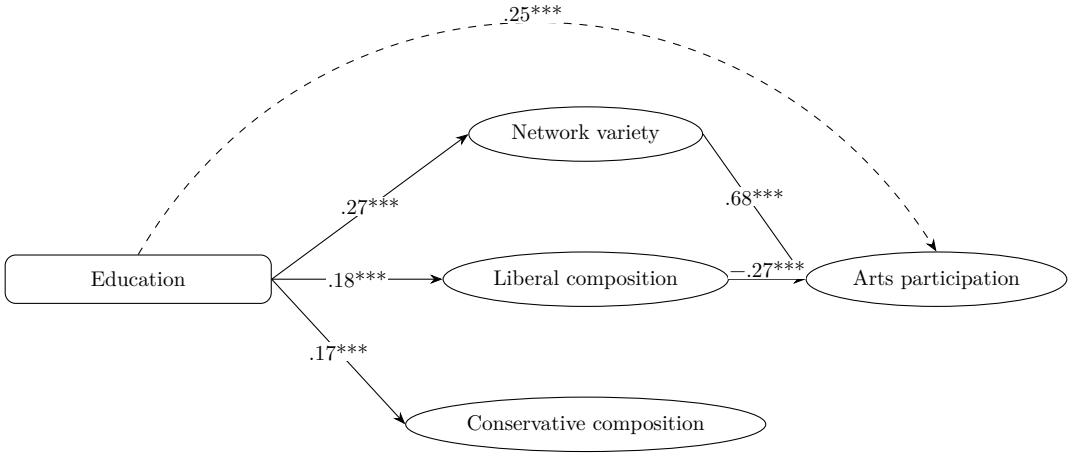
Note. Standardized coefficients from the weak-tie structural model (N = 1,227). Solid arrows are paths through the network factors, dashed arrows are direct paths, and only paths significant at $p < .05$ are drawn. Asterisks follow Table 5.

Figure 7. Paths from Political Conservatism to Arts Participation, Weak Ties



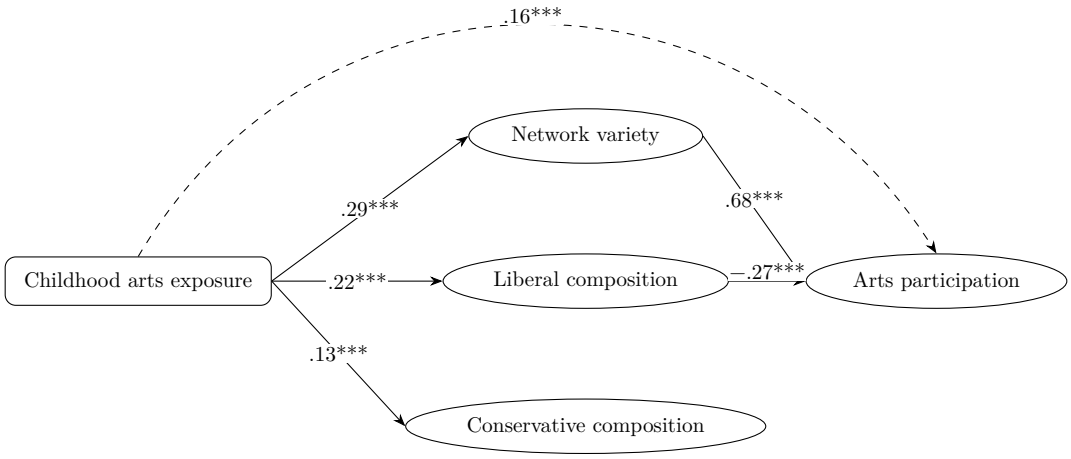
Note. Standardized coefficients from the weak-tie structural model (N = 1,227). Solid arrows are paths through the network factors, dashed arrows are direct paths, and only paths significant at $p < .05$ are drawn. Asterisks follow Table 5.

Figure 8. Paths from Education to Arts Participation, Strong Ties



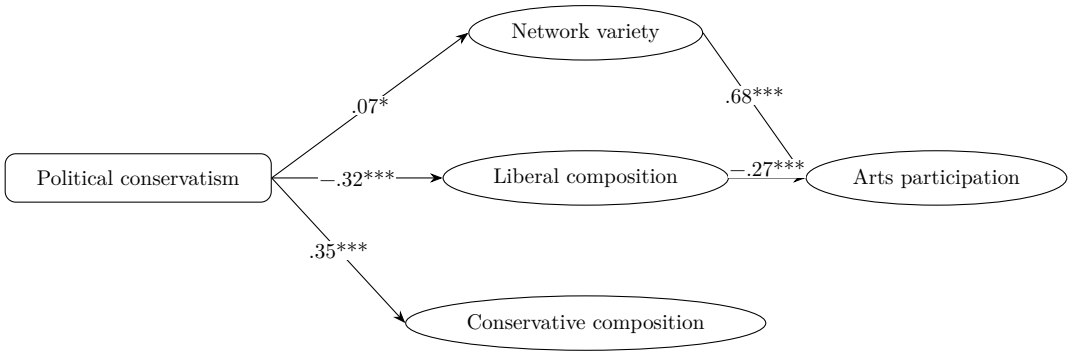
Note. Standardized coefficients from the strong-tie structural model (N = 1,227). Solid arrows are paths through the network factors, dashed arrows are direct paths, and only paths significant at $p < .05$ are drawn. Asterisks follow Table 5.

Figure 9. Paths from Childhood Arts Exposure to Arts Participation, Strong Ties



Note. Standardized coefficients from the strong-tie structural model (N = 1,227). Solid arrows are paths through the network factors, dashed arrows are direct paths, and only paths significant at $p < .05$ are drawn. Asterisks follow Table 5.

Figure 10. Paths from Political Conservatism to Arts Participation, Strong Ties



Note. Standardized coefficients from the strong-tie structural model (N = 1,227). Solid arrows are paths through the network factors, dashed arrows are direct paths, and only paths significant at $p < .05$ are drawn. Asterisks follow Table 5.

Table 10. Direct, Network-Mediated, and Total Standardized Effects of Education, Childhood Arts Exposure, and Political Ideology on the Activity Factors, Weak Ties

Effect	Arts	Solitary leisure	Residual
Education (ordinal)			
Direct	0.20***	0.32***	0.06
Network-mediated	0.18***	0.04	0.20***
Total	0.38***	0.37***	0.26***
Childhood arts exposure			
Direct	0.16***	0.16***	0.07
Network-mediated	0.13***	0.06**	0.12***
Total	0.29***	0.22***	0.19***
Political ideology			
Direct	−0.12*	−0.01	−0.27**
Network-mediated	0.20***	−0.09	0.39***
Total	0.09***	−0.10*	0.12***

Note. Standardized effects from the weak-tie structural model (N = 1,227). Network-mediated effects sum the products of the demographic-to-network and network-to-activity paths across the three weak-tie network factors, and total effects equal direct plus indirect. * p < .05, ** p < .01, *** p < .001, using delta-method standard errors.

Table 11. Direct, Network-Mediated, and Total Standardized Effects of Education, Childhood Arts Exposure, and Political Ideology on the Activity Factors, Strong Ties

Effect	Arts	Solitary leisure	Residual
Education (ordinal)			
Direct	0.25***	0.32***	0.16***
Network-mediated	0.13***	0.05**	0.11***
Total	0.38***	0.37***	0.27***
Childhood arts exposure			
Direct	0.16***	0.16***	0.10*
Network-mediated	0.13***	0.06**	0.09***
Total	0.29***	0.22***	0.19***
Political ideology			
Direct	−0.04	−0.01	−0.10
Network-mediated	0.13***	−0.08	0.22***
Total	0.09***	−0.09*	0.12***

Note. Standardized effects from the strong-tie structural model (N = 1,227). Network-mediated effects sum the products of the demographic-to-network and network-to-activity paths across the three strong-tie network factors, and total effects equal direct plus indirect. * p < .05, ** p < .01, *** p < .001, using delta-method standard errors.

Appendix: Supplementary Measurement Models

Table A1. Oblimin-Rotated Exploratory Factor Loadings for the Activity Battery

Item	Assigned target	Arts	Leisure	Residual
Museum	Arts	0.84	0.08	−0.10
Art gallery	Arts	0.88	0.03	−0.09
Symphony, orchestra, or opera	Arts	0.95	−0.14	−0.11
Carnival, fair, or amusement park	Arts	0.78	−0.02	0.16
Music concert or festival	Arts	0.81	0.01	0.15
Play or musical	Arts	0.86	−0.08	0.00
Library	Arts	0.53	0.19	−0.07
Fancy restaurant	Arts	0.64	0.11	0.29
Dance performance	Arts	0.93	−0.20	−0.02
Movie theater	Arts	0.64	0.05	0.37
Attend sports	Arts	0.73	0.05	0.26
Hiking, camping, or boating	Arts	0.58	0.26	−0.14
Historic site	Arts	0.74	0.12	−0.12
Go to festival	Arts	0.80	0.12	0.01
Go for a walk	Leisure	−0.08	0.79	0.00
Exercise or yoga	Leisure	0.31	0.37	0.25
Read a novel, poem, or play	Leisure	0.35	0.21	−0.11
Gardening	Residual	0.42	0.22	−0.34
Fast food	Residual	0.06	0.06	0.43
Home auto repair	Residual	0.48	0.16	−0.23

Note. Oblimin-rotated pattern loadings from an exploratory factor analysis (WLS estimation, N = 1,243). The EFA fits the 20 items adequately ($\chi^2(133) = 290.8$, $p < .001$, TLI = .911, RMSEA = .080 [.076, .084]). The assigned target column reports the varimax assignment used as the ESEM targets. Factor correlations from the oblique solution are .19 (arts and leisure), .15 (arts and residual), and .01 (leisure and residual).

Table A2. Standardized Loadings and Antonym Residual Covariances from the Separate Weak-Tie and Strong-Tie Confirmatory Models

Item	Weak factor	Loading	Strong factor	Loading
Asian people	Variety	0.83	Variety	0.77
Second home	Variety	0.68	Variety	0.71
Born elsewhere	Variety	0.78	Variety	0.78
Middle Eastern or North African people	Variety	0.84	Variety	0.86
Native Hawaiian or Pacific Islander people	Variety	0.82	Variety	0.83
American Indian people	Variety	0.74	Variety	0.81
LGBTQ people	Liberal	0.69	Liberal	0.71
Little church	Liberal	0.67	Liberal	0.67
Very liberal	Liberal	0.71	Liberal	0.73
Hispanic or Latine people	Liberal	0.77	Variety	0.73
Black people	Liberal	0.76	Variety	0.76
City people	Liberal	0.70	Liberal	0.77
Lots of church	Conservative	0.83	Conservative	0.83
Very conservative	Conservative	0.73	Conservative	0.73
White people	Conservative	0.84	Conservative	0.78
Rural people	Conservative	0.60	Conservative	0.62
Very liberal with very conservative	Residual	−0.32	Residual	−0.55
Little church with lots of church	Residual	0.10	Residual	−0.05
City people with rural people	Residual	−0.40	Residual	−0.44

Note. Standardized loadings from separate three-factor WLSMV confirmatory models for weak ties and strong ties (N = 1,258 each), with the baseline models for the loadings. The residual rows come from the antonym-freed models, and each has a standard error of .02. Gender items are excluded. Item labels are abbreviated from the variable names and should be checked against the question wording.